HDOC Day-54

Question: Write a C program using pointers to find the maximum sum among all contiguous subarrays of length k. Print -1 when k is invalid (k must satisfy 0 < k < size).

C Program:

```
#include <stdio.h>

void max_subarray_sum(int *arr, int size, int k) {
    if (k <= 0 || k >= size) {
        printf("-1\n");
        return;
    }

    int sum = 0;
    for (int i = 0; i < k; i++)
        sum += arr[i];

    int maxSum = sum;
    for (int i = k; i < size; i++) {
        sum += arr[i] - arr[i - k];
        if (sum > maxSum)
            maxSum = sum;
    }
    printf("%d\n", maxSum);
}

int main(void) {
    int size, k;
    scanf("%d", &size);
    if (size <= 0) return 0;
    int arr[size];
    for (int i = 0; i < size; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &k);
    max_subarray_sum(arr, size, k);
    return 0;
}
```

Sample Input and Output:

```
Input:
6
2 1 5 1 3 2
3

Output:
9
```